

SIMATS ENGINEERING

ASSESSMENT TOOL 2

1/4



COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5113

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 100

Annexure A

CONCEPT MAPPING

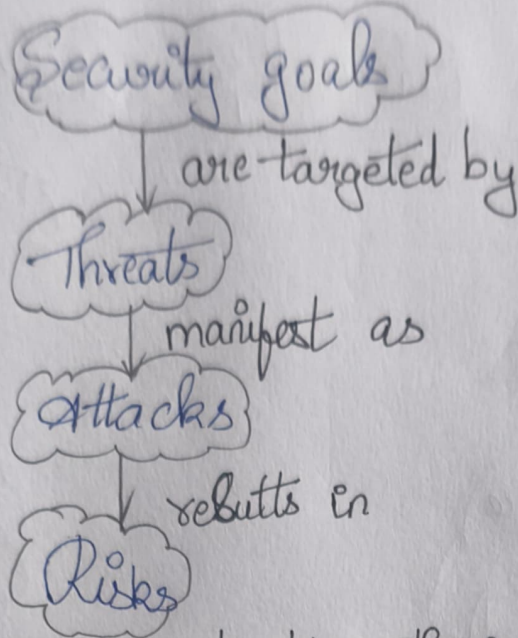
Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

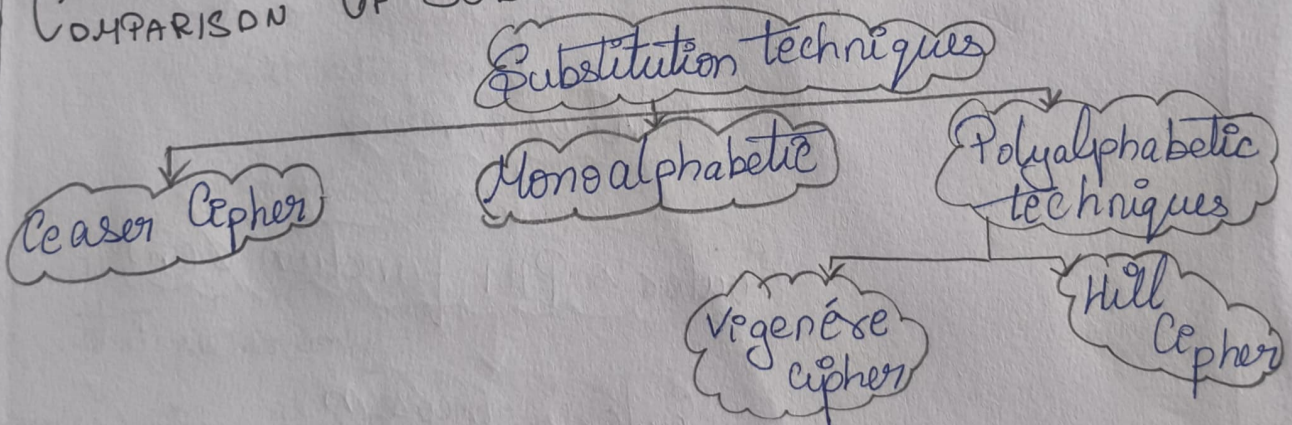
<i>Q. No</i>	<i>Understanding of Concepts (25)</i>	<i>Experimental Design (30)</i>	<i>Use of Tools (20)</i>	<i>Analysis of Results (15)</i>	<i>Documentation (10)</i>	<i>Total Score (out of 100)</i>
<i>1</i>	/ 5	/ 6	/ 4	/ 3	/ 2	
<i>2</i>	/ 5	/ 6	/ 4	/ 3	/ 2	
<i>3</i>	/ 5	/ 6	/ 4	/ 3	/ 2	
<i>4</i>	/ 5	/ 6	/ 4	/ 3	/ 2	
<i>5</i>	/ 5	/ 6	/ 4	/ 3	/ 2	
OVERALL RUBRICS VALUE						
	/ 5	/ 5	/ 4	/ 4	/ 2	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

1) CORE MAP STRUCTURE



* Integrity → Threat: Unauthorized Alteration →
 Attack: Man-in-the-Middle (MitM) → Risk: Fraud
 Mechanism: The attacker secretly relays and alters
 the communication between two parties
 who believe they are directly talking to each other.

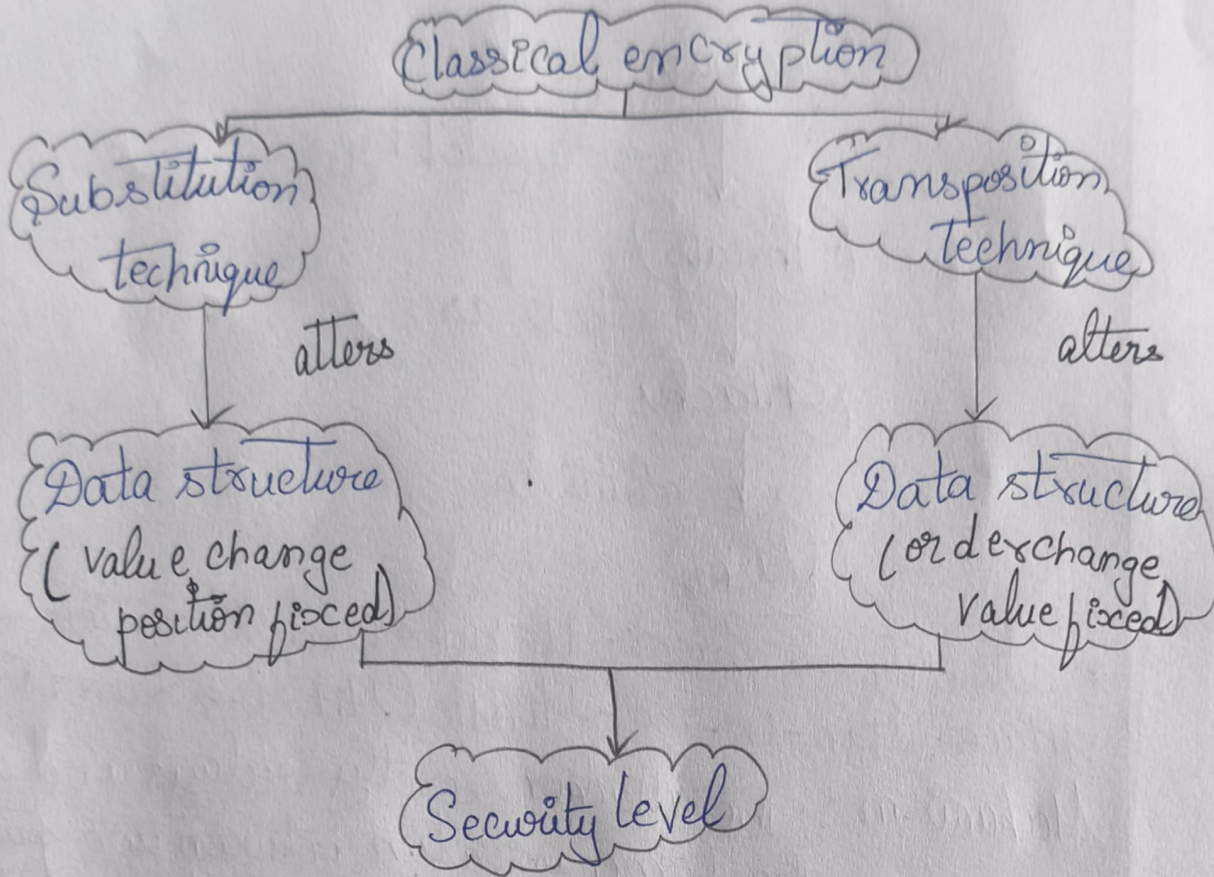
2) COMPARISON OF SUBSTITUTION TECHNIQUES



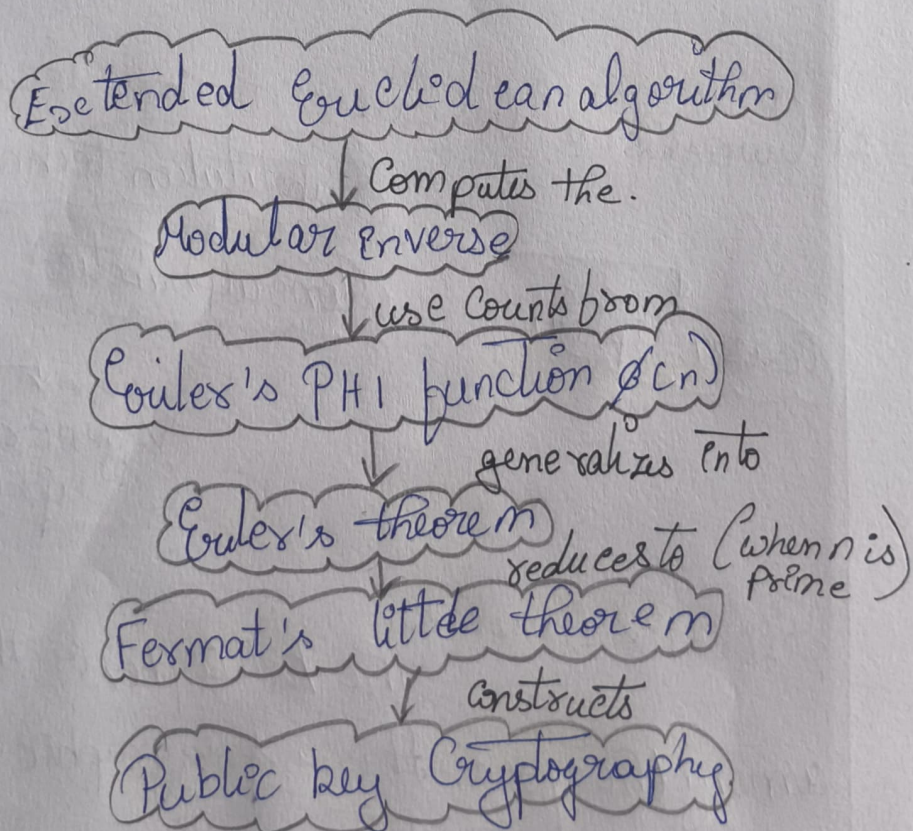
SECURITY ANALYSIS:

The Hill cipher is the most secure among the four. It is a polygraphic cipher, meaning it encrypts multiple letters at the same time.

3) CLASSICAL ENCRYPTION SCHEMES:



5) NUMBER THEORY



4) STEGANOGRAPHY CONCEPTS:

